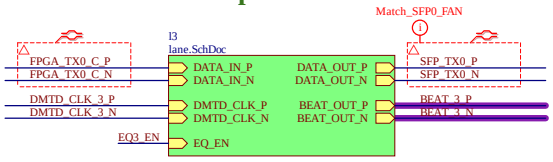
For a new version: 1. add a D-FF to measure the PLL output clock. 2. Remove the phase shifter (does not work on data) and possibly replace it with a working one. 3. Redesign the voltage divider for VREFAC, use a divider per D-FF (see VREFAC note). 4. replace all 0201 and 0102 footprints with larger ones (caps & MOSFETs).

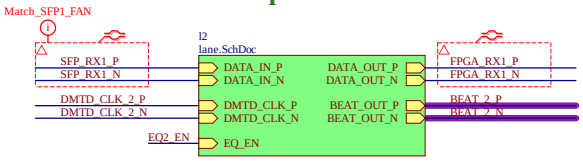
Dedicated to GT 0 (Channel 0) on Quad 0 and SFP 0

UpLink Tx

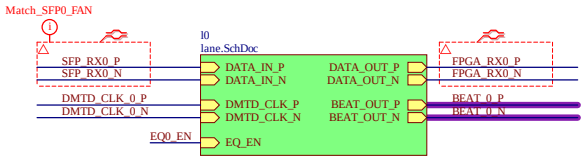


Dedicated to GT 1 (Channel 1) on Quad 0 and SFP 1

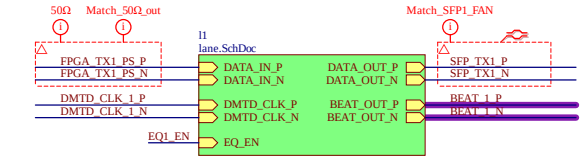
UpLink Rx

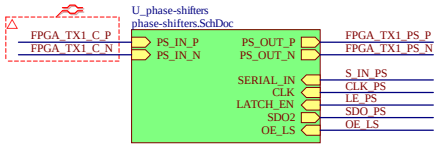


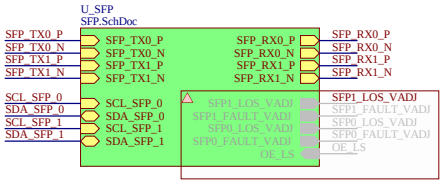
DownLink Rx

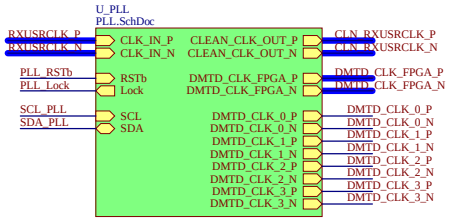


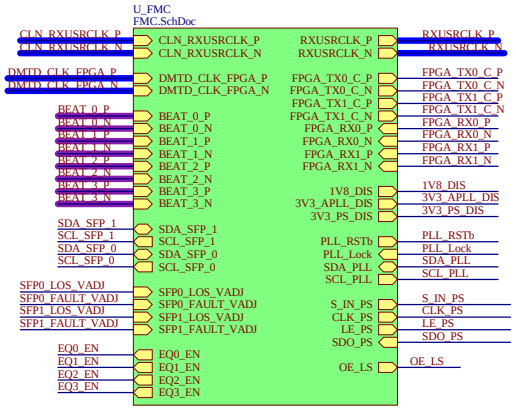
DownLink Tx

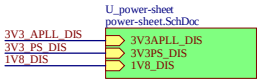








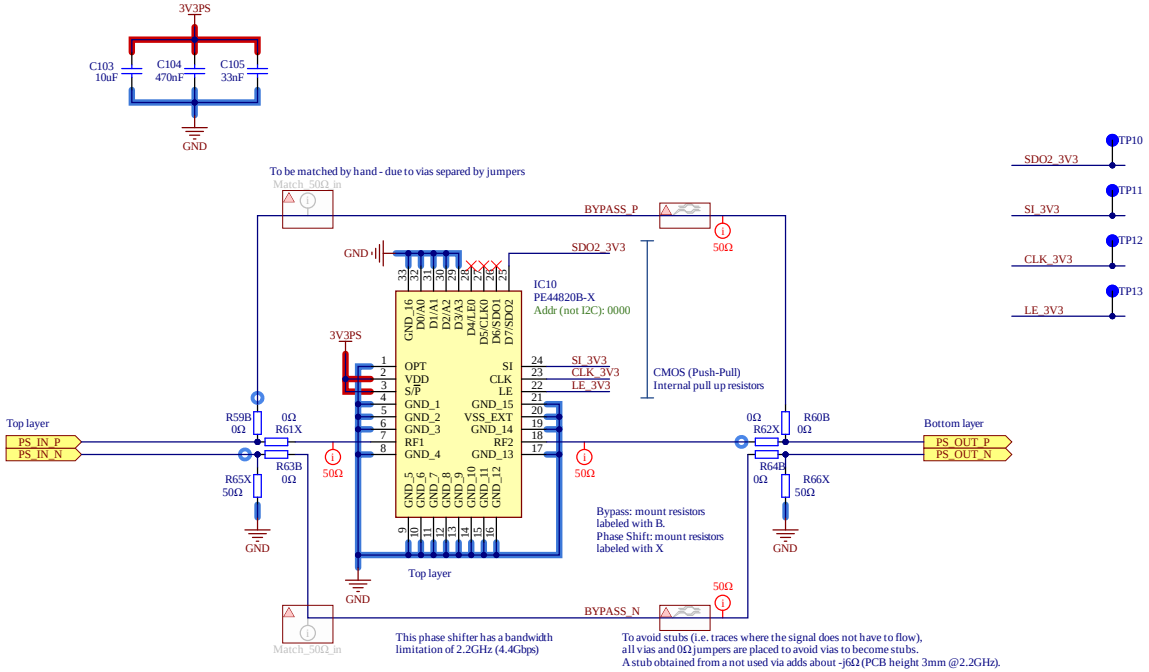


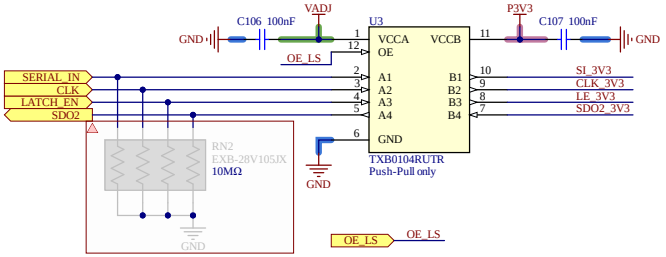


Project Options > Options >

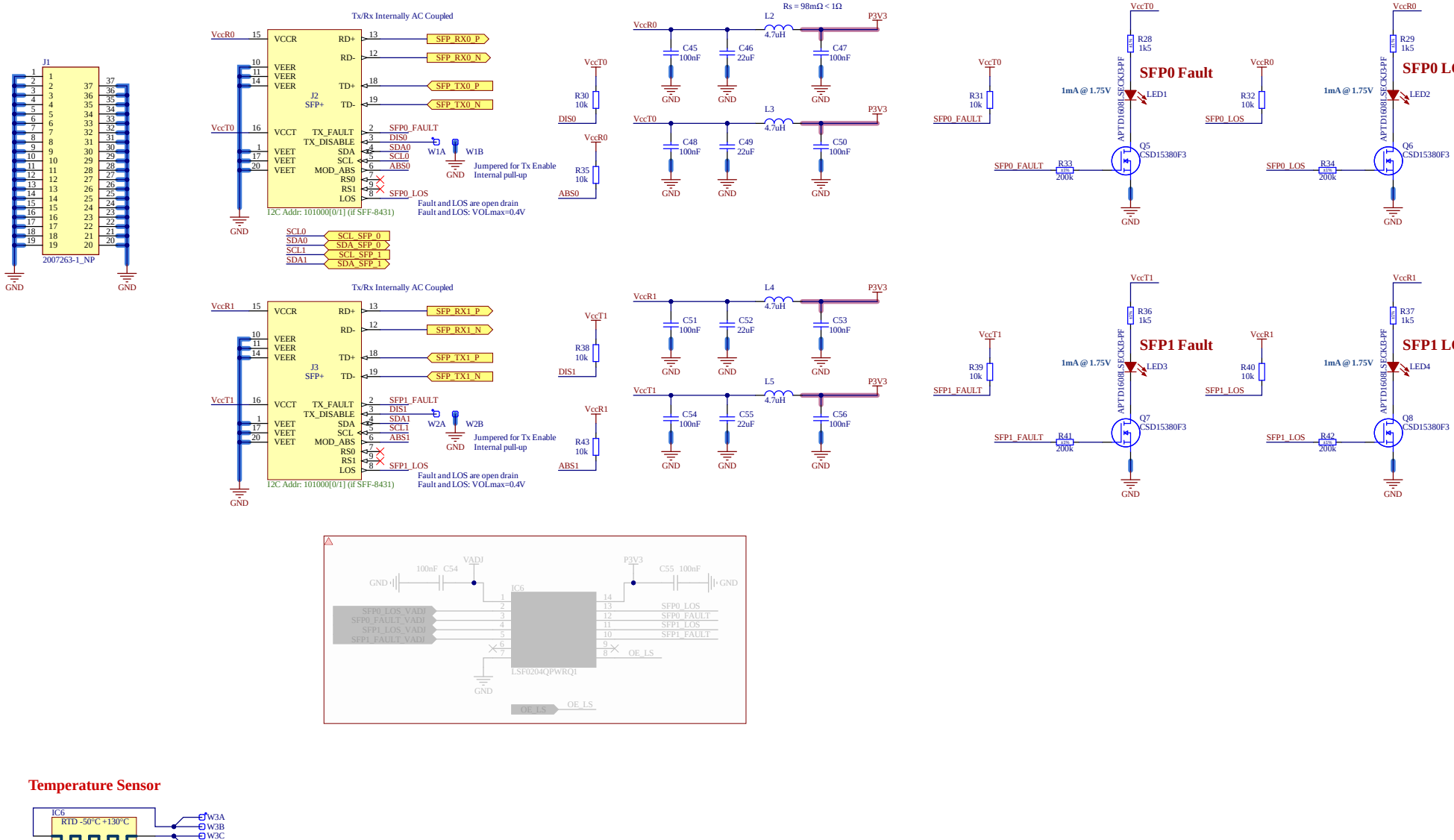
1. Allow ports to name nets (NO)
2. Allow sheet entries to name nets (NO)
3. Append sheet number to local nets (NO)
4. Net identifier scope: Automatic (Hierarchical)
5. Automatic cross reference disabled

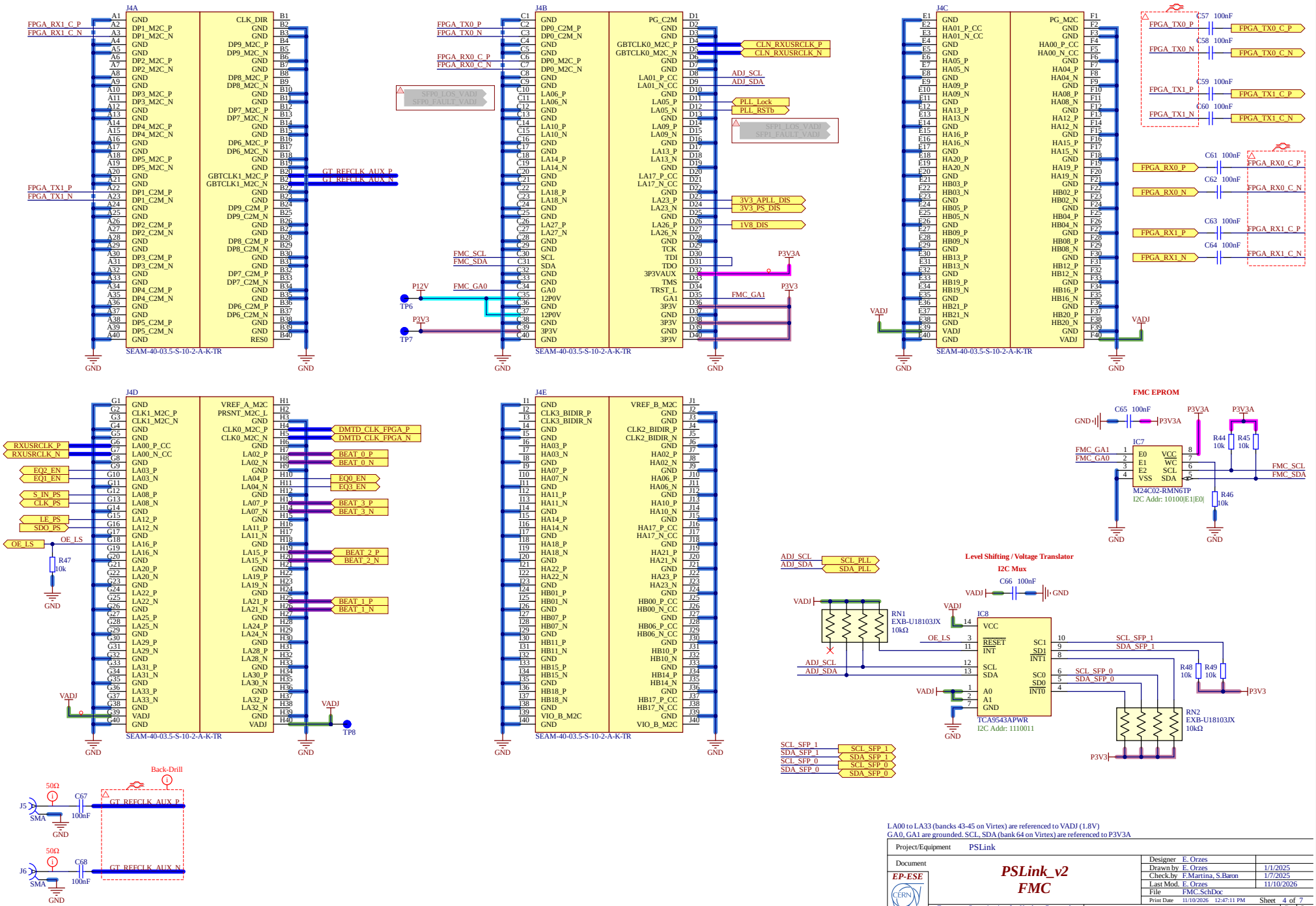
Project/Equipment	PSLink		
Document	**PSLink_v2** **Top**	Designer	E. Orzes
		Drawn by	E. Orzes 1/1/2025
		Check.by	F.Martina, S.Baron 1/7/2025
		Last Mod.	E. Orzes 11/10/2026
		File	top.SchDoc
		Print Date	11/10/2026 12:47:11 PM Sheet 1 of 7
EP-ESE	*European Organization for Nuclear Research* *CH-1211 Genève 23 - Switzerland*	Size	A3
		Rev	1.1





Project/Equipment	PSLink		
Document	**PSLink_v2 Phase Shifter**	Designer	E. Orzes
		Drawn by	E. Orzes 1/1/2025
		Check.by	F.Martina, S.Baron 1/7/2025
		Last Mod.	E. Orzes 11/10/2026
		File	phase-shifters.SchDoc
		Print Date	11/10/2026 12:47:11 PM Sheet 2 of 7
EP-ESE	European Organization for Nuclear Research CH-1211 Genève 23 - Switzerland	Size	A3
		Rev	1.1

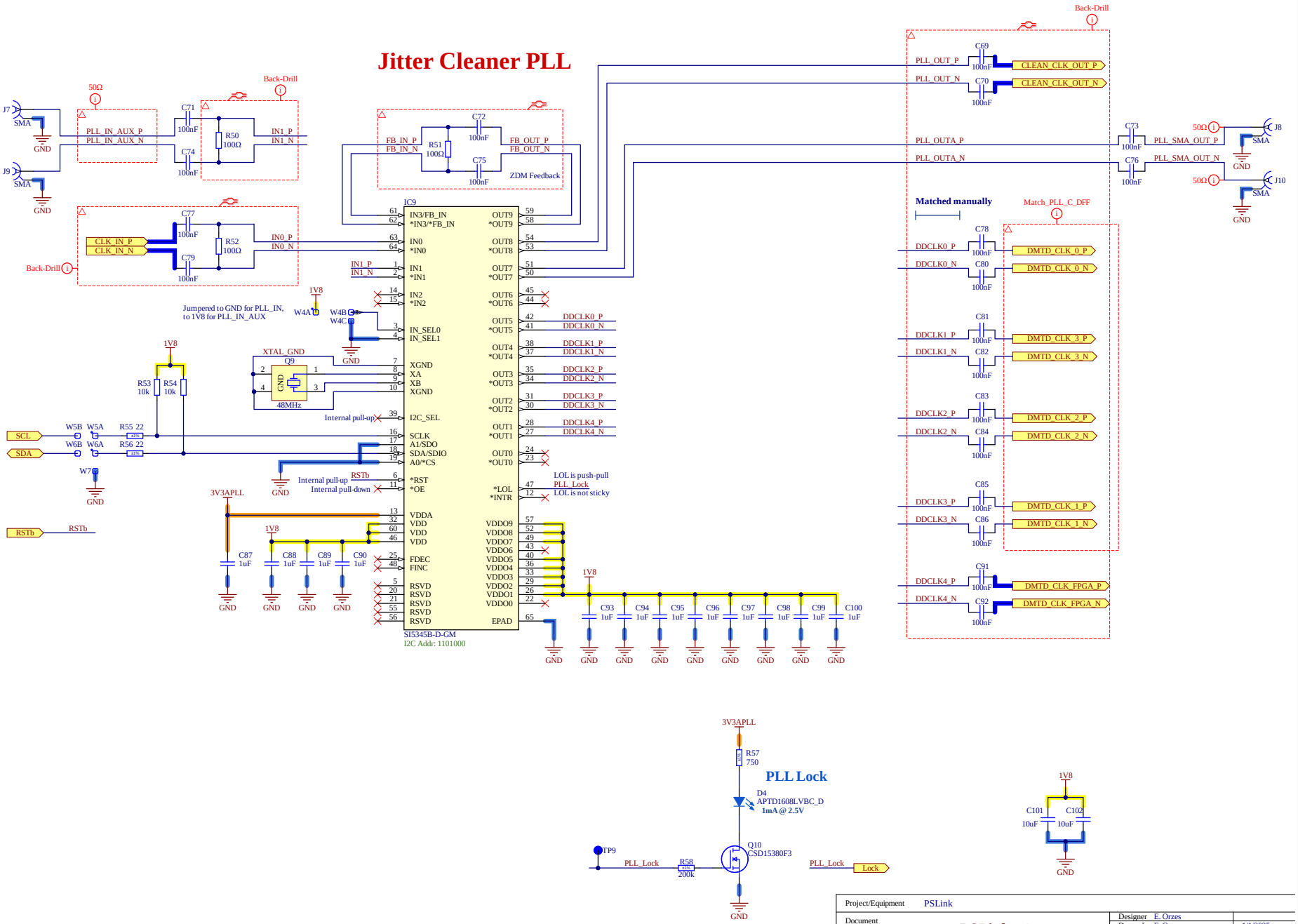

J1
2007263-1_NP
GND
Tx/Rx Internally AC Coupled
VccR0
VccT0
J2
SFP+
VCCR
VEER
VCCT
VEET
RD+
RD-
TD+
TD-
TX_FAULT
TX_DISABLE
SDA
SCL
MOD_ABS
RS0
RS1
LOS
SFP_RX0_P
SFP_RX0_N
SFP_TX0_P
SFP_TX0_N
SFP0_FAULT
DIS0
SDA0
SCL0
ABS0
SFP0_LOS
W1A
W1B
Jumpered for Tx Enable
Internal pull-up
Fault and LOS are open drain
Fault and LOS: VOLmax=0.4V
I2C Addr: 101000[0/1] (if SFF-8431)
R30 10k
R35 10k
SCL0
SDA0
SCL1
SDA1
SCL_SFP_0
SDA_SFP_0
SCL_SFP_1
SDA_SFP_1
Rs = 98mΩ < 1Ω
L2 4.7uH
L3 4.7uH
P3V3
C45 100nF
C46 22uF
C47 100nF
C48 100nF
C49 22uF
C50 100nF
SFP0 Fault
SFP0 LOS
1mA @ 1.75V
LED1
LED2
R28 1k5
R29 1k5
R31 10k
R32 10k
R33 200k
R34 200k
Q5 CSD15380F3
Q6 CSD15380F3
APTD1608LSECK3-PF
VccT1
VccR1
J3
SFP_RX1_P
SFP_RX1_N
SFP_TX1_P
SFP_TX1_N
SFP1_FAULT
DIS1
SDA1
SCL1
ABS1
SFP1_LOS
W2A
W2B
R38 10k
R43 10k
L4 4.7uH
L5 4.7uH
C51 100nF
C52 22uF
C53 100nF
C54 100nF
C55 22uF
C56 100nF
SFP1 Fault
SFP1 LOS
LED3
LED4
R36 1k5
R37 1k5
R39 10k
R40 10k
R41 200k
R42 200k
Q7 CSD15380F3
Q8 CSD15380F3
IC6
VADJ
100nF C54
C55 100nF
SFP0_LOS_VADJ
SFP0_FAULT_VADJ
SFP1_LOS_VADJ
SFP1_FAULT_VADJ
LSF0204QPWRQ1
OE_LS
Temperature Sensor
RTD -50°C +130°C
32207590
W3A
W3B
W3C
W3D
Project/Equipment PSLink
Document
PSLink_v2
SFP
EP-ESE
European Organization for Nuclear Research
CH-1211 Genève 23 - Switzerland
Designer E. Orzes
Drawn by E. Orzes 1/1/2025
Check.by F.Martina, S.Baron 1/7/2025
Last Mod. E. Orzes 11/10/2026
File SFP.SchDoc
Print Date 11/10/2026 12:47:11 PM
Sheet 3 of 7
Size A3
Rev 1.1



LA00 to LA33 (bancks 43-45 on Virtex) are referenced to VADJ (1.8V)
GA0, GA1 are grounded. SCL, SDA (bank 64 on Virtex) are referenced to P3V3A

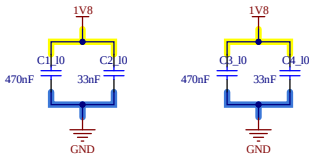
Project/Equipment	PSLink		
Document	**PSLink_v2 FMC**	Designer	E. Orzes
EP-ESE		Drawn by	E. Orzes 1/1/2025
		Check.by	F.Martina, S.Baron 1/7/2025
		Last Mod.	E. Orzes 11/10/2026
		File	FMC_SchDoc
		Print Date	11/10/2026 12:47:11 PM
		Sheet	4 of 7
European Organization for Nuclear Research *CH-1211 Genève 23 - Switzerland*		Size	A3
		Rev	1.1

Jitter Cleaner PLL

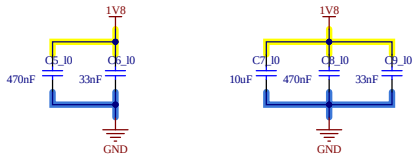


Project/Equipment	PSLink		
Document	PSLink_v2 PLL	Designer E. Orzes	
		Drawn by E. Orzes	1/1/2025
		Check.by F.Martina, S.Baron	1/7/2025
		Last Mod. E. Orzes	11/10/2026
		File PLL.SchDoc	
		Print Date 11/10/2026 12:47:12 PM	Sheet 5 of 7
EP-ESE	European Organization for Nuclear Research CH-1211 Genève 23 - Switzerland		Size A3 / Rev 1.1

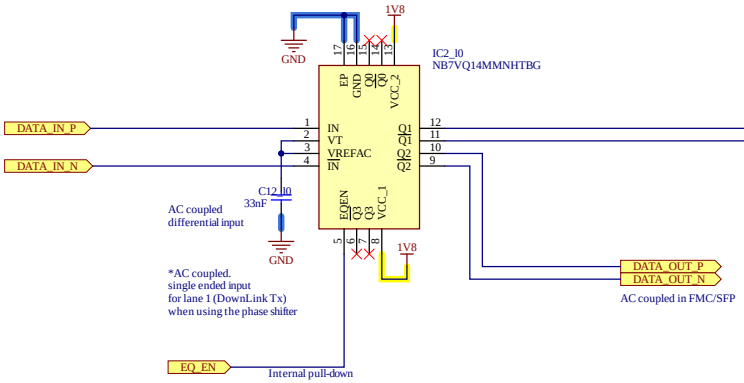
Bypass for Fanout power pins



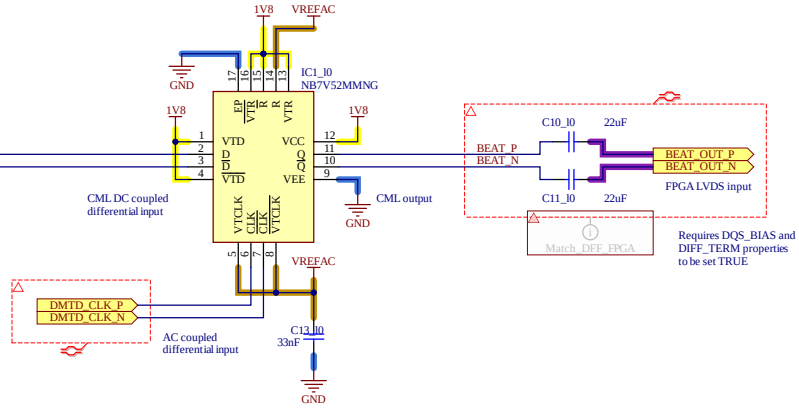
Bypass for D-FF power pins



Fanout Buffer

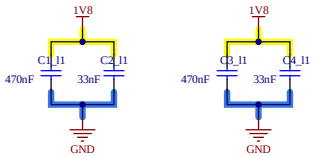


D-FF

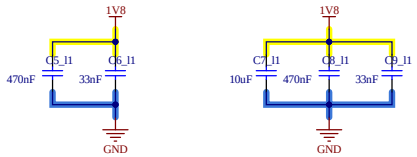


Project/Equipment	PSLink		
Document	PSLink_v2 Lane	Designer E. Orzes	
EP-ESE		Drawn by E. Orzes	1/1/2025
		Check.by F.Martina, S.Baron	1/7/2025
		Last Mod. E. Orzes	11/10/2026
		File lane.SchDoc	
		Print Date 11/10/2026 12:47:12 PM	Sheet 6 of 7
European Organization for Nuclear Research CH-1211 Genève 23 - Switzerland			Size A3 / Rev 1.1

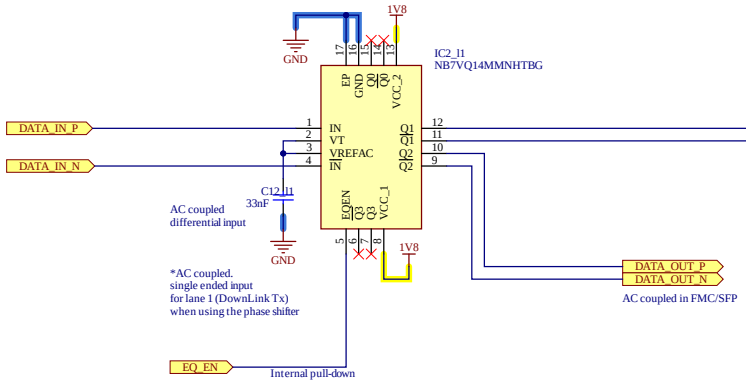
Bypass for Fanout power pins



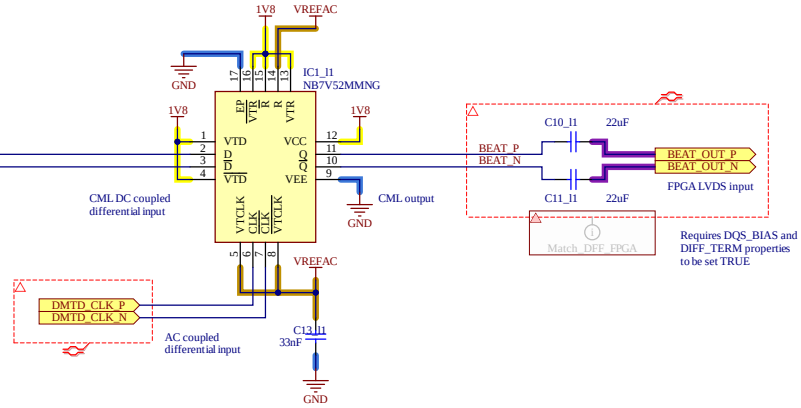
Bypass for D-FF power pins



Fanout Buffer

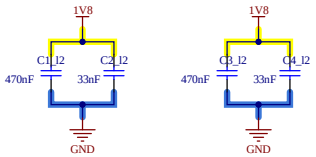


D-FF

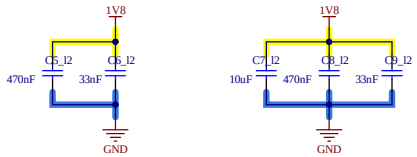


Project/Equipment	PSLink		
Document	*PSLink_v2 Lane*	Designer E. Orzes	
		Drawn by E. Orzes	1/1/2025
		Check.by F.Martina, S.Baron	1/7/2025
		Last Mod. E. Orzes	11/10/2026
		File lane.SchDoc	
		Print Date 11/10/2026 12:47:12 PM	Sheet 6 of 7
EP-ESE	*European Organization for Nuclear Research* *CH-1211 Genève 23 - Switzerland*		Size A3 / Rev 1.1

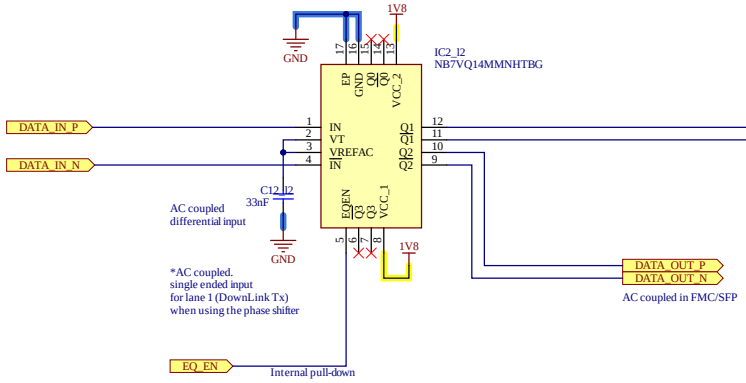
Bypass for Fanout power pins



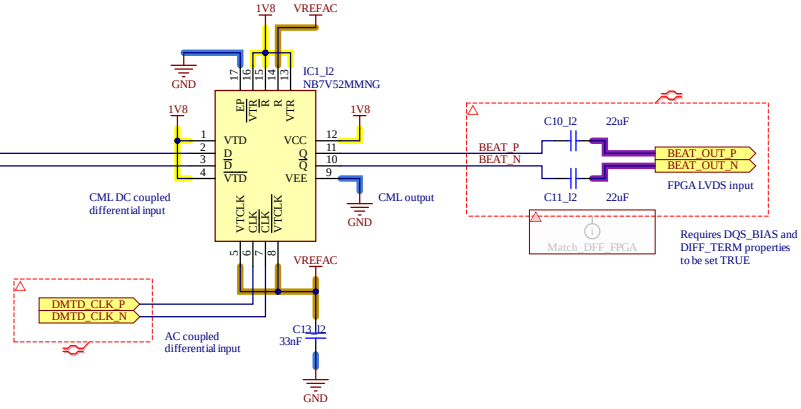
Bypass for D-FF power pins



Fanout Buffer

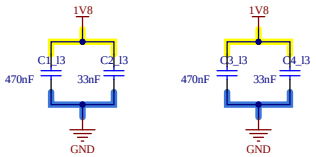


D-FF

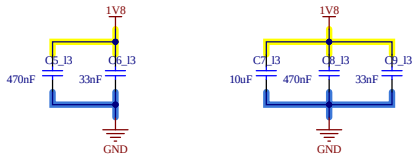


Project/Equipment	PSLink		
Document	PSLink_v2 Lane	Designer	E. Orzes
		Drawn by	E. Orzes 1/1/2025
		Check.by	F.Martina, S.Baron 1/7/2025
		Last Mod.	E. Orzes 11/10/2026
		File	lane.SchDoc
		Print Date	11/10/2026 12:47:12 PM Sheet 6 of 7
EP-ESE	European Organization for Nuclear Research CH-1211 Genève 23 - Switzerland	Size	A3
		Rev	1.1

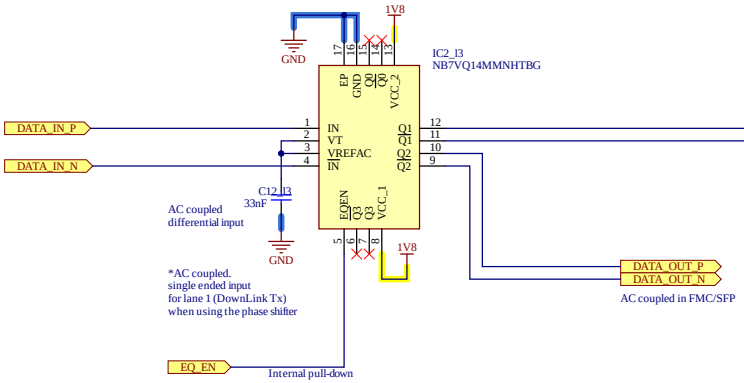
Bypass for Fanout power pins



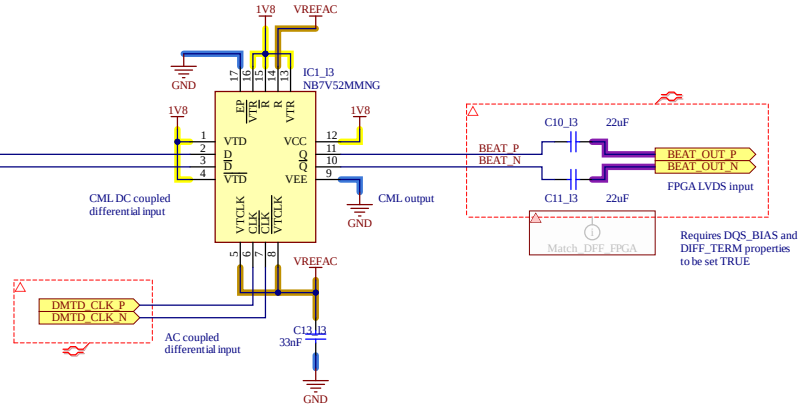
Bypass for D-FF power pins



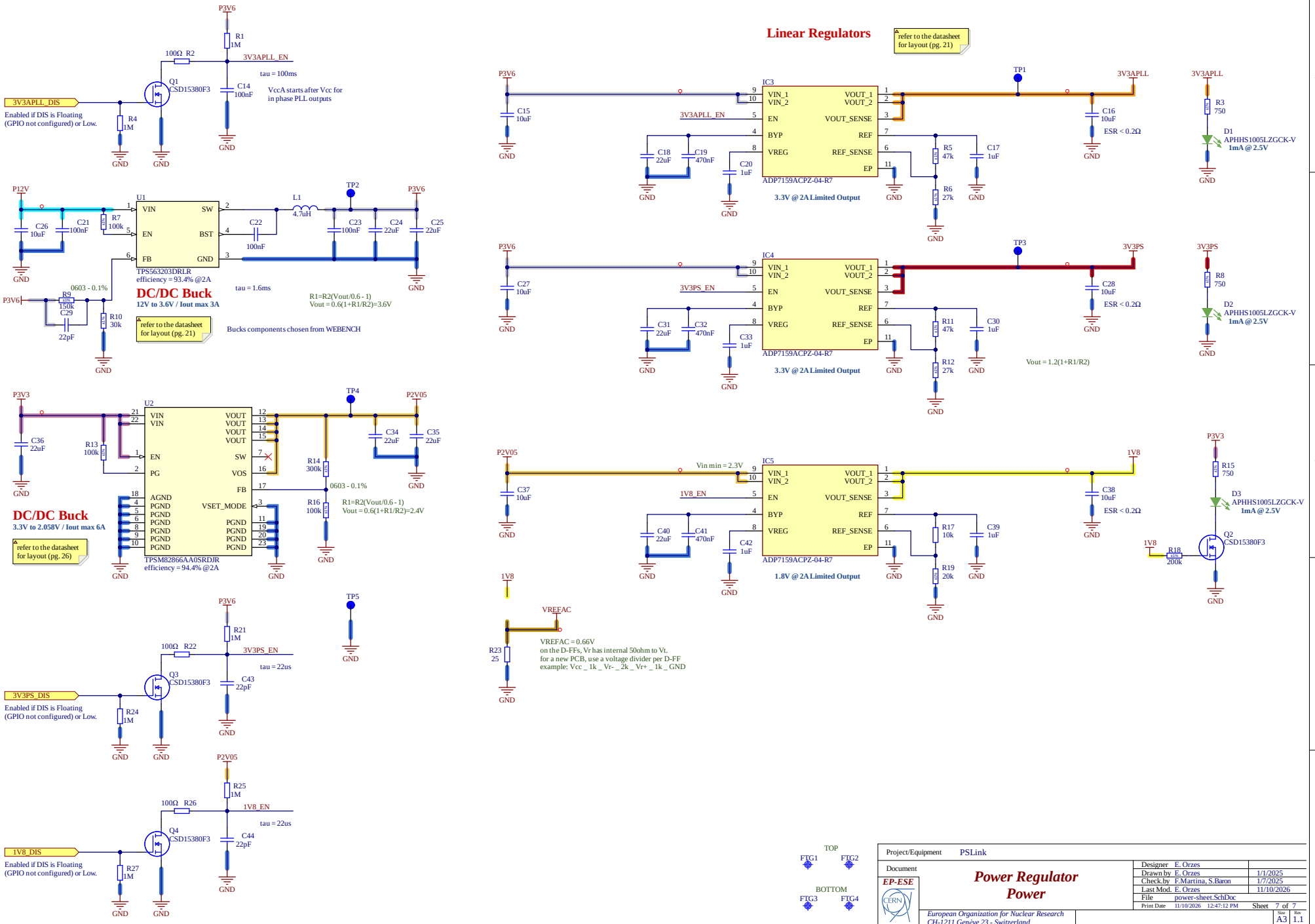
Fanout Buffer



D-FF



Project/Equipment	PSLink		
Document	PSLink_v2 Lane	Designer E. Orzes	
		Drawn by E. Orzes	1/1/2025
		Check.by F.Martina, S.Baron	1/7/2025
		Last Mod. E. Orzes	11/10/2026
		File lane.SchDoc	
		Print Date 11/10/2026 12:47:12 PM	Sheet 6 of 7
EP-ESE	European Organization for Nuclear Research CH-1211 Genève 23 - Switzerland		Size A3 / Rev 1.1


Linear Regulators
refer to the datasheet for layout (pg. 21)
P3V6
R1 1M
100Ω R2
3V3APLL_EN
Q1 CSD15380F3
tau = 100ms
C14 100nF
VccA starts after Vcc for in phase PLL outputs
3V3APLL_DIS
Enabled if DIS is Floating (GPIO not configured) or Low.
R4 1M
GND
P12V
U1
VIN
SW
EN
BST
FB
GND
TPS563203DRLR
efficiency = 93.4% @2A
L1 4.7uH
TP2
C26 10uF
C21 100nF
R7 100k
C22 100nF
C23 100nF
C24 22uF
C25 22uF
DC/DC Buck
12V to 3.6V / Iout max 3A
tau = 1.6ms
R1=R2(Vout/0.6 - 1)
Vout = 0.6(1+R1/R2)=3.6V
R9 150k 0603 - 0.1%
C29 22pF
R10 30k
refer to the datasheet for layout (pg. 21)
Bucks components chosen from WEBENCH
P3V3
U2
VIN
EN
PG
AGND
PGND
VOUT
SW
VOS
FB
VSET_MODE
TPSM82866AA0SRDJR
efficiency = 94.4% @2A
TP4
P2V05
C36 22uF
R13 100k
C34 22uF
C35 22uF
R14 300k
0603 - 0.1%
R16 100k
R1=R2(Vout/0.6 - 1)
Vout = 0.6(1+R1/R2)=2.4V
DC/DC Buck
3.3V to 2.058V / Iout max 6A
refer to the datasheet for layout (pg. 26)
TP5
R21 1M
100Ω R22
3V3PS_EN
Q3 CSD15380F3
tau = 22us
C43 22pF
3V3PS_DIS
Enabled if DIS is Floating (GPIO not configured) or Low.
R24 1M
R25 1M
100Ω R26
1V8_EN
Q4 CSD15380F3
tau = 22us
C44 22pF
1V8_DIS
Enabled if DIS is Floating (GPIO not configured) or Low.
R27 1M
IC3
IC4
IC5
VIN_1
VIN_2
VOUT_1
VOUT_2
EN
VOUT_SENSE
BYP
REF
VREG
REF_SENSE
EP
ADP7159ACPZ-04-R7
3.3V @ 2A Limited Output
1.8V @ 2A Limited Output
C15 10uF
C18 22uF
C19 470nF
C20 1uF
R5 47k
R6 27k
C17 1uF
C16 10uF
ESR < 0.2Ω
TP1
3V3APLL
R3 750
D1 APHHS1005LZGCK-V
1mA @ 2.5V
C27 10uF
C31 22uF
C32 470nF
C33 1uF
R11 47k
R12 27k
C30 1uF
C28 10uF
TP3
3V3PS
R8 750
D2 APHHS1005LZGCK-V
1mA @ 2.5V
Vout = 1.2(1+R1/R2)
C37 10uF
C40 22uF
C41 470nF
C42 1uF
R17 10k
R19 20k
C39 1uF
C38 10uF
1V8
R15 750
D3 APHHS1005LZGCK-V
1mA @ 2.5V
Q2 CSD15380F3
R18 200k
Vin min = 2.3V
1V8_EN
VREFAC
R23 25
VREFAC = 0.66V
on the D-FFs, Vr has internal 50ohm to Vt.
for a new PCB, use a voltage divider per D-FF
example: Vcc _ 1k _ Vr- _ 2k _ Vr+ _ 1k _ GND
TOP
FTG1
FTG2
BOTTOM
FTG3
FTG4
Project/Equipment PSLink
Document
Power Regulator
Power
EP-ESE
CERN
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